Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0041407867494825	1 – 1.0045379537953796

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0099791109716103	1.0052981990439502 – 1.0114145589878802

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.000220482857458	0.9982380795066622 – 1.0015236833388539

Matrix multiplication

256 × 256 · 256 × 256 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.006558129554634	1.0040813729657436 – 1.0097556013671134

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0022646283129366	1.0001415744780857 – 1.0036768129396727

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9219211161792327	0.9046610169491526 – 0.9351600651112317

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0073907695598696	1.0036293584230918 – 1.0111569141817014

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0028731220528566	0.9951774102038144 – 1.0099328267811916

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

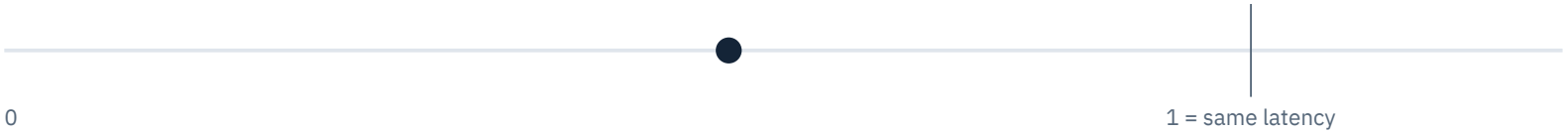
Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.659749670619236	0.6589557171183079 – 0.6666666666666666

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5811209439528023	0.5733855185909981 – 0.5885518590998043

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9937565403473292	0.9928150613248039 – 1.0013695787507781

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.994797745689799	0.9879273217853035 – 1.0047720257365078

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9983466021802162	0.9966300720587722 – 1.002325631779009

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0052951143744704	0.9939748392112516 – 1.0154761476794765

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.009035905294692	1.0026501946355793 – 1.0426416803806298

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9993552131020697	0.9974357330598115 – 1.0067443286327409